

CONEPRO · REQUIREMENTS

ConePro requirements

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DOCUMENT CONTROL

Document ID	CNP-REQ-001	Version	0.3
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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/conepro		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from CNP-CAL-001; decisions in CNP-DDR-001 (R6 met by D1, R9 assumes the lever of D2, new R15 from D3); R10 total defined as carried in the bag
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

ConePro requirements

These requirements were checked by calculation at TRL 3 in CNP-CAL-001 against the design decided in CNP-DDR-001. Seven are met on paper, two are at risk, two are not met and four cannot be verified until hardware exists. The misses are **R5**, whose 1,000 mm clause needs an extension rod that is not in the BOM, and **R10**, where the instrument carried in its bag weighs about 16.7 kg against 16 kg. The two at risk are R2 (depth accuracy) and R11 (shock survival). Targets are still proposals for review, not yet validated with users, and will be revised after co-design (see CNP-PRB-001). No target was relaxed at TRL 3; R15 is new and carries decision D3.

Reference test. A single test point driven to 850 mm in a medium-stiff soil averaging 15 mm per blow (about 57 blows), by one person, on level ground.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status at TRL 3 (CNP-CAL-001)	Verification (later)
R1	Standard DCP geometry	8.0 kg hammer, 575 mm free drop, 16 mm rod, 20 mm cone with a 60 degree point, replaceable cone; tolerances as ASTM D6951	Met (nominal): model gives 8.00 kg and 575 mm; tolerances to check against the standard	Drawing check against the standard
R2	Depth measurement	Resolution 0.5 mm or finer; error within ± 1 mm over a 1,000 mm range compared with a steel rule	At risk: 0.046 mm resolution; bench error ± 0.98 mm worst case, ± 0.49 mm root sum square; a 2 degree lean adds up to 1.8 mm at 850 mm after tilt correction	Bench calibration
R3	Blow detection	Every blow counted and no false counts from handling, over a 200-blow sequence	Not verifiable at TRL 3: Hall margin 2.3 times at the worst angle; three-signal logic (Hall, accelerometer peak, depth)	Bench blow sequence
R4	Per-blow record	Depth after each blow logged within 0.5 s of impact, with blow number and time	Met (calculation): settled within 36 ms, recorded at 0.21 s	Firmware sketch review
R5	Penetration range	850 mm or more per lower rod; 1,000 mm or more with one extension rod and automatic re-zero	Not met (1,000 mm clause): 938 mm stroke, 850 mm usable, but no extension rod in the BOM; re-zero unverified	Model check; field test
R6	Verticality	Warn the operator when the rod is more than 5 degrees off vertical	Met by design (D1): accelerometer reads tilt to about ± 0.3 degree; bubble level as backup	Bench tilt check
R7	Phone output	Live plot of depth against blows; DCP index per layer; indicative CBR from the ASTM D6951 correlations with the soil type selected; CSV export with time, location and operator notes	Not verifiable at TRL 3 (app specified, not written)	App specification review
R8	Works offline	No cell signal needed; logger stores 50 or more tests if the phone is absent or disconnects	Met (calculation): about 830 tests in 4 MB	Firmware sketch review
R9	One-person test time	Reference test, including setup and rod extraction with the optional lever (D2), in 12 min or less by one person	Met (estimate), thin: 8.1 to 11.7 min	Timed trials
R10	Portability	Total mass as carried in the bag, cells included and the optional lever excluded, 16 kg or less; heaviest piece 10 kg or less; packed length 1.1 m or less	Not met (total): 16.7 kg; heaviest piece 9.8 kg and packed length about 1.08 m are met	Mass roll-up; weighing
R11	Rugged electronics	Logger, sensors and connectors IP65; operate at 0 to 45 °C; anvil-mounted parts survive 10,000 blows	At risk: about 1,400 to 5,400 g mean at the anvil; about 530 g on the isolated pad against 10,000 g part ratings; cable and connector fatigue not calculated	Drop-count test

ID	Requirement	Target	Status at TRL 3 (CNP-CAL-001)	Verification (later)
R12	Battery life	8 h or more of continuous logging on user-replaceable non-lithium cells	Met (calculation): about 40 h, about 20 h at 0 °C	Power measurement
R13	Cost	\$400 or less in parts for one complete instrument (optional lever excluded)	Met: \$321; \$371 with the lever	Priced BOM (bom/ bom.csv)
R14	Honest results	App states the correlation used and that results are indicative and not for foundation design	Not verifiable at TRL 3 (app not written)	App specification review
R15	Retrofit fit (D3)	Sensor set (BOM items 8 to 12) fits a standard ASTM D6951 penetrometer with a 16 mm rod and a 50 to 80 mm anvil, without machining	Not verifiable at TRL 3: interfaces sized in the model; needs a commercial DCP to check	Fit check on a commercial DCP

■ Assumptions

- Soil behavior, blow rate and extraction time are estimates, not measurements. A practiced operator drops the hammer about once every 2.5 s; a slow one every 3.5 s.
- CBR correlations are those given in ASTM D6951 (from US Army Corps of Engineers work); other soils and regions may need local correlations.
- Mass estimates use steel at 7,850 kg/m³ and aluminum at 2,700 kg/m³ applied to the parametric model `cad/src/model.py` , plus catalog masses for bought parts (CNP-CAL-001, Table 2).
- Battery life assumes an average electronics draw of about 50 mA and about 2,000 mAh usable from AA alkaline cells at that current, halved at 0 °C.
- All other calculation assumptions are listed in CNP-CAL-001, Table 1.